

Clinical and Pedigree Evaluation of Familial Developmental Impairment

1. Pedigree and Clinical Analysis

The pedigree showed unaffected parents with affected children, suggesting an autosomal recessive (AR) inheritance pattern. Affected individuals were present in both sexes, and no consanguineous marriages were reported within the families. Since all families originated from the same village, an autosomal recessive inheritance pattern was considered the most likely mode of inheritance.

Clinically, the affected individuals presented with learning difficulties, mild developmental delay, and intellectual disability. These clinical features were subsequently used to prioritize genes and variants potentially associated with neurodevelopmental disorders.

2. Variant Filtering Workflow

An annotated VCF file was provided for variant analysis. To identify candidate variants, a population frequency threshold of 0.05 (5%) was initially applied as the filtering criterion. Variants with a frequency greater than 0.05 were excluded based on the frequency information available in the 1000 Genomes, ExAC, and ESP annotation columns.

This threshold was retained as the predefined filtering criterion for this analysis. Next, synonymous variants were excluded, while variants located in exonic and splicing regions were retained for further investigation.

The remaining variants were then categorized according to their genotype and allele fraction (AF):

Heterozygous variants: allele fraction approximately 0.5

Homozygous variants: allele fraction approximately 1.0

The homozygous variants were of particular interest because the pedigree suggested a possible autosomal recessive mode of inheritance.

Note: The 0.05 frequency threshold was used as the predefined criterion in this analysis. For rare Mendelian disorders, stricter population-frequency thresholds are often applied during variant prioritization, depending on the disease model and phenotype.

3. Gene Panel and Variant Annotation

A phenotype-oriented candidate gene panel was created in a separate Excel worksheet. Genes potentially associated with the patients' clinical features, particularly developmental delay and intellectual disability, were included.

The genes corresponding to the filtered heterozygous and homozygous variants were then added to the same worksheet. Duplicate variants were identified to facilitate subsequent variant review and prioritization. Candidate genes were evaluated based on their known disease associations, inheritance patterns, reported phenotypes, variant characteristics, and available clinical evidence.

4. Clinical Significance and Variant Validation

The final candidate variants were evaluated using multiple genomic and clinical databases, including:

- gnomAD – population allele frequency
- ClinVar – clinical interpretation of genetic variants
- OMIM – gene–disease and phenotype relationships
- GeneCards – gene function and disease associations
- DECIPHER – genotype–phenotype relationships
- Franklin – clinical variant interpretation and ACMG-based evidence
- Iranome – population variation data from Iranian individuals

Evaluation criteria

Candidate variants were assessed according to the following criteria:

A) Population frequency

Variants were evaluated for their frequency in population databases.

B) Clinical classification

Available classifications were reviewed, including:

Pathogenic

Likely Pathogenic

Variant of Uncertain Significance (VUS)

Likely Benign

Benign

C) Phenotype matching

Reported phenotypes associated with each candidate gene were compared with the clinical features observed in the affected individuals.

D) Inheritance compatibility

The reported inheritance pattern of the associated disorder was compared with the suspected inheritance pattern in the family.

E) Variant characteristics

Variant type, zygosity, sequencing quality, predicted molecular consequence, and available functional or clinical evidence were considered.

5. Evaluation of Candidate Gene: FDFT1

5.1 Gene Overview

FDFT1 (*farnesyl-diphosphate farnesyltransferase 1*) is a protein-coding gene located on **chromosome 8p23.1**. It encodes squalene synthase, an enzyme involved in the cholesterol biosynthesis pathway. Squalene synthase catalyzes the conversion of two molecules of farnesyl diphosphate (FPP) to squalene, representing the first committed step in cholesterol biosynthesis. (Database, n.d.)

The FDFT1 protein is approximately 416 amino acids in length and is associated with the endoplasmic reticulum. The protein contains an N-terminal catalytic domain and a C-terminal membrane-spanning region involved in its association with the endoplasmic

reticulum. Coman D, Vissers L, Waterham H, et al. Squalene Synthase Deficiency. (GeneReviews®. University of Washington)

5.2 Disease Association

Biallelic pathogenic variants in FDFT1 are associated with squalene synthase deficiency (SQSD), a rare autosomal recessive inborn error of cholesterol biosynthesis. The condition has multisystem manifestations and has been reported to include developmental delay/intellectual disability, seizures, structural brain abnormalities, cortical visual impairment, optic nerve hypoplasia, photosensitivity, and other congenital abnormalities. The molecular diagnosis of SQSD can be established by identifying biallelic pathogenic or likely pathogenic variants in FDFT1, although characteristic biochemical abnormalities can also support the diagnosis. Importantly, the identification of biallelic variants of uncertain significance (VUS) alone is not sufficient to establish the diagnosis. The reported inheritance pattern of SQSD is autosomal recessive, which is compatible with the suspected inheritance pattern in the family analyzed in this study. (GeneReviews®. University of Washington)

5.3 Variant Evaluation

A homozygous in-frame deletion, NM_001287750:c.193_198del (p.65_66del), was identified in FDFT1.

The variant showed high sequencing quality in the analyzed sample, with a high read depth and genotype quality. The variant was also observed at a very low population frequency, supporting its rarity.

At the protein level, p.65_66del represents an in-frame deletion of two amino acids rather than a frameshift variant. Therefore, the deletion is not predicted to disrupt the reading frame of the protein.

Because the variant is located near the N-terminal portion of the protein, it is important to consider its potential effect on protein structure and function. The established catalytic activity of squalene synthase is associated with the N-terminal catalytic domain, while the C-terminal region contains the membrane-spanning domain. Therefore, the precise functional impact of the p.65_66del deletion cannot be determined solely from its position and requires additional evidence. (GeneReviews®. University of Washington)

5.4 Clinical and Population Evidence

The identified variant was considered potentially relevant because it was:

- Homozygous, consistent with the suspected AR inheritance pattern
- Rare in population databases
- Located in a protein-coding region
- Predicted to cause an in-frame deletion
- Located in a gene with an established association with an autosomal recessive neurodevelopmental/metabolic disorder

However, rarity and homozygosity alone are not sufficient to establish pathogenicity. The available evidence does not currently provide sufficient support to classify p.65_66del as pathogenic. In particular, additional evidence regarding the specific variant, including functional studies, independent clinical observations, and segregation within the family, would strengthen the interpretation.

5.5 Overall Assessment

FDFT1 was prioritized as a candidate gene because its established autosomal recessive disease association is compatible with the suspected inheritance pattern in the family, and the clinical spectrum of squalene synthase deficiency includes developmental delay and intellectual disability. (GeneReviews®. University of Washington)

The identified homozygous p.65_66del variant is rare and technically well supported in the analyzed sample. However, because it is an in-frame deletion and there is insufficient variant-specific evidence to establish a pathogenic effect, it should currently be considered a candidate variant rather than a confirmed disease-causing variant.

Further investigation, particularly segregation analysis, functional studies, and additional clinical/phenotypic correlation, would be required to determine whether this variant contributes to the disease phenotype.

6. Conclusion

The bioinformatics analysis identified several candidate variants and genes potentially associated with the observed neurodevelopmental phenotype. Candidate prioritization was based on population frequency, variant consequence, zygosity, sequencing quality, clinical database annotations, gene–disease associations, inheritance patterns, and phenotype matching.

Among the evaluated candidates, FDFT1 was considered noteworthy because biallelic pathogenic variants in this gene are associated with squalene synthase deficiency, an autosomal recessive disorder that can include developmental delay and intellectual disability. (GeneReviews®. University of Washington)

Nevertheless, the identified FDFT1 c.193_198del (p.65_66del) variant cannot currently be considered definitively pathogenic based on the available evidence. Additional experimental and familial evidence will be necessary for further assessment.

References

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